

AGRiTRUST: A Provenance-Grounded Benchmark for Bengali Agricultural Retrieval

Khan Raiyan Ibne Reza
raiyen.reza@northsouth.edu
North South University
Dhaka, Bangladesh

Sanjana Aktar Maria
sanjana.maria@northsouth.edu
North South University
Dhaka, Bangladesh

Sumaiya Tabassum Nimi
sumaiya.nimi@northsouth.edu
North South University
Dhaka, Bangladesh

Abstract

We introduce AGRiTRUST, a provenance-grounded resource for Bengali agricultural retrieval, built from 284 authoritative publications issued by five Bangladeshi government and research organizations. The collection comprises a 2,882-node knowledge graph and 1,000 queries (900 answerable), with gold-node mappings verified by three independent annotators (Fleiss’ $\kappa=0.72$). Each retrieval unit is a provenance-preserving knowledge node spanning natural-language content, structured facts, and deterministic source metadata, supporting strict document-level gold matching.

We evaluate five retrieval architectures and six embedding models across three controlled language conditions. BM25 is the strongest single retriever in Bengali ($R@10=0.506$), moderately ahead of a correctly configured general-purpose dense embedding model ($R@10=0.464$); Reciprocal Rank Fusion of the two reaches the highest overall accuracy ($R@10=0.539$). This aggregate picture hides a sharp query-register effect: dense retrieval nearly fails on colloquial farmer queries ($R@10=0.093$) yet reaches $R@10=0.970$ on formal, safety-critical queries, a gap invisible to unstratified evaluation.

This pattern reverses when queries cross the script boundary. BM25 collapses to near-random performance under cross-lingual querying ($R@10=0.004$), while dense embeddings largely hold up ($R@10=0.425$). We also find that a single, easily overlooked embedding API parameter changes reported $R@10$ by 7 $\times$, indicating that configuration auditing should precede any architecture comparison in low-resource retrieval settings. The benchmark, knowledge graph, and evaluation harness are available at <https://huggingface.co/datasets/RaiyanKhaan/AgriTrust-RAG>.

Keywords

Bengali, low-resource retrieval, RAG, agricultural advisory, benchmark, dense retrieval, BM25, hybrid fusion, cross-lingual retrieval, multilingual evaluation, embedding configuration

1 Introduction

1.1 Motivation and Gap

Retrieval-Augmented Generation (RAG) systems are increasingly deployed in domain-specific settings, yet evaluated almost exclusively on English benchmarks. Bengali agricultural advisory presents a natural stress test: with 237 million speakers [2], advisory queries use colloquial farmer language while authoritative source documents use formal scientific Bengali. This register gap challenges lexical and dense retrieval in opposing ways. Of these, 200 queries cover safety-critical content (pesticide dosage, off-label chemical use), where retrieval failure is a real-world risk, not merely a benchmark penalty.

Existing retrieval benchmarks (BEIR [20], CRAG [23]) are English-only; multilingual collections like MIRACL [27] and Mr. TyDi [26] do not target agricultural advisory or query-register variation. Prior Bengali RAG systems translate queries before retrieval [6] or do not isolate the retrieval layer [12], while the closest agricultural resource [15] is not a retrieval-only benchmark with provenance-level gold labels or controlled cross-lingual conditions.

1.2 The AGRiTRUST Collection

Unlike existing retrieval benchmarks that pair text chunks with aggregate accuracy scores, AGRiTRUST is built around six properties that jointly define its contribution: provenance-grounded retrieval units, structured canonical nodes rather than raw text, full document traceability, a controlled multilingual protocol, agricultural domain grounding, and register-aware evaluation covering both farmer queries and formal queries. AGRiTRUST consists of: (i) 284 source PDFs from five Bangladeshi government and research organizations; (ii) 2,882 knowledge nodes (1,022 image-linked), each a provenance-preserving retrieval unit; (iii) a knowledge graph of 19,768 entities and 17,501 factual triples; (iv) 1,000 queries (900 answerable) across three categories (farmer-anchored, KG-grounded, safety), with gold-node mappings verified by three independent annotators (Fleiss’ $\kappa=0.72$); and (v) an evaluation harness implementing provenance-level gold matching, bootstrap

confidence intervals, and an embedding API configuration audit.

These four findings (summarized in the abstract) are developed in full in Section 5, with mechanistic explanations in Section 6.

1.3 Contributions

1. **AGRiTRUST benchmark collection.** A provenance-grounded test collection for low-resource Bengali agricultural retrieval, covering 284 authoritative documents, 2,882 knowledge nodes, 19,768 entities, 17,501 triples, and 1,000 annotated queries (available at <https://huggingface.co/datasets/RaiyanKhaan/AgriTrust-RAG>).
2. **Canonical node representation.** A layered retrieval unit uniting natural-language content, structured facts, and deterministic source metadata via bounded, schema-guided extraction.
3. **Evaluation harness & audit protocol.** A reproducible framework implementing provenance-level gold matching, bootstrap confidence intervals, and an embedding API configuration audit.
4. **Benchmark validation via systematic diagnosis.** Evaluation across five architectures, six embedding models, and three language conditions demonstrates that AGRiTRUST’s register- and script-stratified design surfaces failure modes invisible to aggregate, single-condition evaluation.

2 Related Work

2.1 IR and RAG Evaluation Benchmarks

Standard RAG evaluation benchmarks such as BEIR [20], RAGBench [3], GaRAGe [18], and CRAG [23] are built almost entirely in English; recent extensions like T²-RAGBench [19] target text-and-table retrieval in financial documents rather than provenance-grounded, low-resource domain advisory. Multilingual retrieval benchmarks such as MIRACL [27] (18 languages) and Mr. TyDi [26] extend evaluation to diverse languages, but none enable controlled architecture-isolated retrieval evaluation with provenance grounding in a low-resource agricultural setting. Standard benchmarks typically emphasize aggregate accuracy scores, concealing subgroup failure across query registers, a limitation highlighted by concurrent work on stratified retrieval evaluation [10].

2.2 Retrieval in Low-Resource Languages

Dense retrieval often degrades outside high-resource languages due to tokenization fragmentation and language bias [5, 11, 24]. Cross-lingual DPR shows modest gains [22], while multi-vector late-interaction models such as ColBERT [9] improve fine-grained token matching.

Table 1: Positioning of AGRiTRUST relative to closest prior benchmarks and systems.

Resource	Low-res. lang.	Arch. isolated	Retr. only	Cross- lingual	Prov. ground.
BEIR [20]	✗	✓	✓	✗	✗
CRAG [23]	✗	✗	✗	✗	✗
MIRACL [27]	✓	✗	✓	✓	✗
Mr. TyDi [26]	✓	✗	✓	✓	✗
Amharic study [24]	✓	✗	✓	✗	✗
Cross-lingual RAG [6]	✓	✗	✗	✓	✗
Prior advisory benchmark [?]	✓	✗	✗	✗	✓
KinyaColBERT [13]	✓	✗	✓	✗	✗
AGRiTRUST (this work)	✓	✓	✓	✓	✓

In agricultural RAG, KinyaColBERT [13] attributes low accuracy in Kinyarwanda to vocabulary coverage over architecture choice. Query-document vocabulary mismatch is an established retrieval challenge across domains, from general human-system communication [4] and cross-genre retrieval [7] to medical consumer health search [25], where lay queries match formal domain text.

In Bengali, BenLLM-Eval [8] and TigerLLM [14] evaluate LLM capabilities, while KrishokBondhu [1] automates voice advisory and Farmer.Chat [17] extends AI-powered advisory to smallholder farmers across multiple languages. Closest to our setting, a prior advisory resource [?] provides a QA generation dataset, and Hossain et al. [6] translate queries before retrieval. Unlike generation-focused efforts, AGRiTRUST establishes a retrieval-only benchmark to isolate retriever architecture performance under controlled register and language shifts.

As summarized in Table 1, AGRiTRUST is among the first benchmarks to combine a low-resource language, architecture-isolated evaluation, retrieval-only measurement, controlled cross-lingual conditions, and provenance-grounded gold labels in a single design. For example, while MIRACL and Mr. TyDi provide diverse multilingual datasets, their standard evaluations do not isolate retrieval architectures across controlled intra-language register shifts (marked ✗ for Arch. isolated).

3 The AGRiTRUST Benchmark

We construct AGRiTRUST around three requirements (Figure 1): (i) document-level provenance, so every retrieval decision traces to a single source; (ii) knowledge-graph grounding, so queries are verified against structured facts instead of free text; and (iii) independent verification of node quality and query-document mappings, not gold labels accepted from a single automated pass.

3.1 Source Materials

We collect 284 Bengali agricultural PDFs from five government and research organizations (BRRI, IRRI, DAE, SRDI, MoA). The Ministry of Agriculture (MoA) and its extension arm (DAE) jointly contribute approximately 65% of the corpus via national farming handbooks, while specialized institutes (BRRI, IRRI, SRDI) supply targeted diagnostic matrices. Many source documents contain scanned pages, complex layouts, tables, and embedded figures. We therefore convert each PDF into page-level Markdown using a layout-preserving OCR pipeline based on Mistral OCR, retaining document structure and page provenance for subsequent node construction. The resulting Markdown corpus (2,680 section-level passages) serves solely as an intermediate pre-processing artifact. While 73% of this underlying raw text originates from a prior QA benchmark [?], AGRiTRUST’s primary contribution is restructuring this text into a novel, provenance-grounded knowledge graph (100% of KG entity and triple construction, node schema, and the full 900-query gold-mapping are new; only the underlying text and colloquial queries are inherited), establishing a dedicated retrieval benchmark where prior work left retrieval for future evaluation.

Corpus Diversity. To ensure robust retrieval evaluation across the agricultural long tail, the extracted corpus spans 25 years of publications (1999 to 2024) encompassing five major document genres: national farming handbooks, diagnostic manuals, pesticide whitelists, recommendation cards, and extension guides. The resulting entity graph explicitly indexes 915 unique crops (spanning horticulture, agroforestry, and aquaculture), 704 disease variants, and 2,729 chemical or pesticide entities.

3.2 Canonical Knowledge Node Construction

Each knowledge node is a provenance-preserving retrieval unit spanning three layers: natural-language content, structured agricultural facts, and provenance (source, page, organization). Generation is bounded, not open-ended: extraction is constrained to information explicitly present in the source text, and provenance fields are injected deterministically, not model-generated. A representative node is given in Appendix B; the full extraction and generation prompts are detailed in Appendix D.

Deterministic Provenance Injection. Citation metadata (publisher, source document, page range) is never LLM-generated: the generation prompt structurally forbids the model from producing citation fields, which are instead injected by a separate deterministic script directly from markdown trace comments. This decouples provenance

from semantic generation, making citation hallucination (fabricated page numbers or source names) structurally impossible rather than merely unlikely. A separate post-hoc verification pass cross-checks every extracted crop and entity mention against the source markdown text, flagging any node whose entities do not appear in its cited passage for correction.

Chunking. We segment text into 2,882 topic-coherent knowledge nodes, each corresponding to a single agricultural concept. Nodes average $\approx 1,180$ characters, reflecting natural concept boundaries rather than fixed token windows, and are distributed across a 13-category agricultural taxonomy, ranging from Variety (695 nodes) and Cultivation Practice (570) down to a long tail such as Food Safety (18).

Entity and Triple Extraction. We extract 19,768 entities (6.9/node) and 17,501 factual triples via schema-guided, bounded open information extraction. Per-node entity coverage exceeds 96%.

Source-Grounded Automated Verification. Node content (summary, symptoms, management, entities) was generated by Gemini-3.1-Flash-Lite from the source Markdown into a structured JSON schema. Each generated node was then independently verified by a separate model, GPT-5-Nano, which received *both* the node and its ground-truth source passage and scored the pair (0–5) along four criteria: faithfulness (every claim traceable to the source text), completeness, hallucination (unsupported content not present in source), and crop/entity relevance. Using a separate model for generation and verification reduces self-confirmation bias. Because the verifier sees the source passage, not just the node in isolation, its check is constrained to factual consistency and never becomes an open-ended quality rating. Across the resulting graph, entity coverage is 96.1% (2,771 of 2,882 nodes contain at least one extracted entity) and factual triple coverage reaches 100%.

Closed-Loop Refinement. Nodes falling below threshold (≤ 3 on any dimension; 283 of 2,882 nodes, 9.8%) were returned to Gemini-3.1-Flash-Lite together with the verifier’s specific error trace and regenerated against the same source passage, then re-verified. This repeated until each node passed threshold, yielding the final 2,882-node collection without discarding agricultural content.

Human Audit. Because the automated stage already enforces source-level faithfulness, human review targeted a layer the automated check is not designed to catch: domain-appropriate terminology and information completeness under compression, judgments requiring agricultural expertise

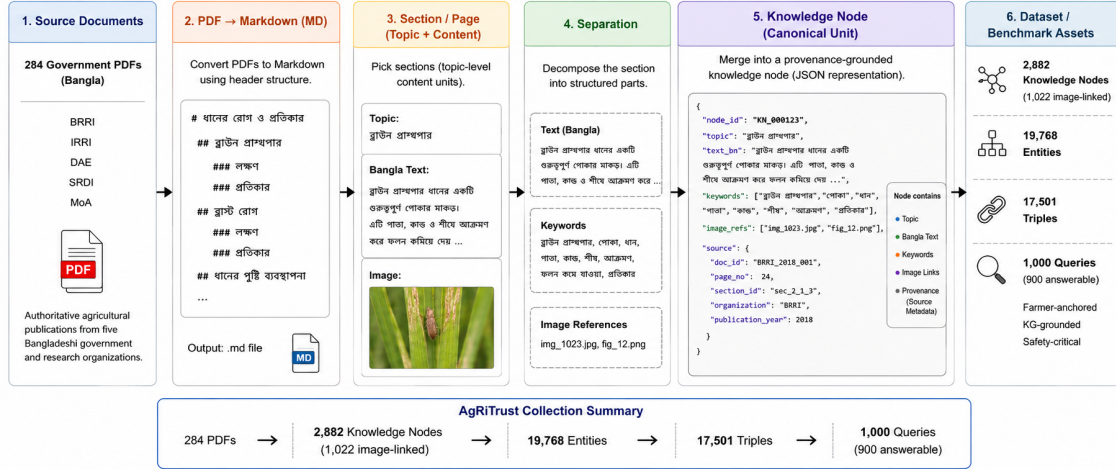


Figure 1: Canonical knowledge node construction pipeline. Source documents are parsed, chunked by topic, and processed into structured, provenance-grounded JSON nodes.

beyond simple text-source comparison. Three expert agricultural annotators independently evaluated a stratified random sample of 200 nodes on two dimensions: information completeness (critical dosage/withholding details dropped under compression) and technical terminology (formal Bengali agricultural terms vs. code-mixed transliterations). Annotators reached strong agreement (Fleiss’ $\kappa=0.81$); 12 of 200 nodes (6.0%) fell below consensus threshold and were corrected via direct manual edit against the source Markdown. Each node remains linked to its originating Markdown passage through deterministic provenance metadata while image-containing nodes continue through the multimodal extension described in §3.3.

3.3 Image-Linked Nodes

Of the 2,882 canonical knowledge nodes, 1,022 (35.5%) additionally carry a linked visual asset (a diagnostic photograph, variety chart, or table-image) captured via an `image_refs` field on the same text node, under the identical provenance schema (publisher, source_document, source_pages) as the node’s textual content. Each image reference additionally carries a figure-type label (one of 12 categories, e.g., *symptom_close_up*, *variety_portrait*, *procedure_illustration*) and a bilingual visual description (`visual_description_en/bn`) generated by a vision-language model, textualizing symptom color, shape, and texture rather than requiring a dedicated vision encoder at retrieval time. These fields extend the schema in Appendix C but, like the image reference itself, are not separately indexed or evaluated in the present retrieval experiments (§5).

This connective structure is motivated directly by the register gap identified in §6.1. Because farmer queries disproportionately describe visually observable symptoms (“leaves turning yellow”) instead of the formal entity names (*Tungro virus*) used in source documents, image-linked nodes are the natural mechanism by which a future system could close exactly the severe retrieval gap this benchmark exposes ($R@10=0.093$ on farmer queries), rather than acting as a general-purpose multimodal extension. We report their structure in Appendix C and leave dedicated visual retrieval to future work (§7.2).

3.4 Query Benchmark

The benchmark contains 1,000 queries across three categories (Table 2).

Farmer-anchored. 400 farmer-anchored queries adapted from a prior QA benchmark [?]. After three-annotator independent gold-node mapping, 300 achieved majority-vote consensus and are retained as answerable; the remaining 100 lacked majority-vote agreement among annotators (Fleiss’ κ for this subset: <0.4) and are excluded from retrieval evaluation. These 100 excluded queries constitute the *low-agreement* subset; disagreement may reflect ambiguous queries or multiple valid gold documents rather than strict unanswerability.

KG-grounded (400). Newly constructed multi-hop questions derived from KG triples. We note a labeling asymmetry: while formal and colloquial queries underwent independent 3-way generation, the KG-grounded queries were derived from single-expert assignments followed by 3-way

verification. Manual inspection of 60 early failure cases refined the gold-matching criterion before any architecture was evaluated, preserving the blind-evaluation guarantee.

Safety (200). All 200 queries test advisory failure modes (pesticide dosage, off-label chemical use, dangerous agromomic advice), each with a verified gold document. Safety queries use formal, precise language (chemical product codes, BRRI variety names) that closely mirrors document vocabulary: they show $1.6\times$ higher lexical overlap with their gold document than farmer queries (§6.1), consistent with their high retrievability in Section 5.3. All 200 are answerable and included in the 900-query evaluation set.

Independent Verification. Farmer-anchored and safety queries were mapped by three annotators (Fleiss’ $\kappa = 0.72$, $n=600$). Majority vote determines the gold node; queries without majority agreement are excluded. KG-grounded gold nodes (400 queries) were assigned by a domain expert annotator and independently validated by three native Bengali-speaking annotators who evaluated each assignment on factual grounding, faithfulness, and relevance (Fleiss’ $\kappa=0.78$, $n=400$). The final benchmark contains 900 answerable queries (300 farmer + 400 KG + 200 safety); the 100 excluded farmer queries are withheld as a low-agreement subset.

3.5 Cross-Lingual Conditions

To isolate language as an explanatory variable independent of retrieval architecture, we translate both the full query set and all 2,882 KG nodes into English, guided by a domain-specific agricultural glossary (312 domain-critical terms, cross-referenced against NCBI Taxonomy; 97% back-translation equivalence on a 100-query spot check) to preserve technical terminology. This produces three controlled conditions on an otherwise identical benchmark: Bengali queries against the Bengali KB (BN→BN), English queries against the Bengali KB (EN→BN, cross-lingual), and English queries against the translated English KB (EN→EN). Because all conditions share the same corpus content and evaluation code, any performance difference is attributable to language.

4 Experimental Design

We evaluate five architectures and six embeddings not to advance retrieval methodology, but because architecture-isolated evaluation is itself part of the benchmark’s contribution (§2.1). Our evaluation is designed to isolate whether retrieval behavior is primarily determined by architecture, embedding model, or language condition. This exhaustive approach demonstrates that AGRiTRUST surfaces failure modes that remain invisible to single-architecture or

Table 2: AGRiTRUST benchmark statistics.

Property	Value
Source PDFs	284
Organizations	5 (BRRI, IRRI, DAE, SRDI, MoA)
KG nodes	2,882
Image-linked nodes	1,022 (35.5%)
Unique entities	19,768
Factual triples	17,501
Queries (total generated)	1,000
Farmer-anchored (answerable)	300
Farmer-anchored (low-agreement)	100
KG-grounded (answerable)	400
Safety-critical (answerable)	200
Answerable evaluation set	900
Inter-annotator κ (farmer+safety)	0.72
Inter-annotator κ (KG-grounded)	0.78
Entity coverage	>96%

aggregate-only evaluations. We organize primary experiments around four research questions: which architecture leads in Bengali (RQ1); whether this depends on the language boundary (RQ2); whether query type moderates performance (RQ3); and how embedding choice affects results (RQ4). We additionally conduct a configuration audit to verify that architecture comparisons are not artifacts of implementation settings. Mechanistic explanations are in §6.

4.1 Retrieval Architectures

We evaluate five representative retrieval systems spanning four retrieval families: (i) **BM25** [16] uses Okapi weighting ($k_1=1.5$, $b=0.75$) with character bigram plus full-word tokenization; (ii) **Dense (Gemini)** uses L_2 -normalized 3,072-dim gemini-embedding-001 vectors with exact inner-product search (FAISS IndexFlatIP) and asymmetric task types; (iii) **Dense (BGE-M3 Native)** uses 1,024-dim native dense embeddings as an open, retrieval-specific baseline; (iv) **ColBERT** [9] applies BGE-M3 multi-vector scoring at 512-token passage granularity (to match the dense baselines’ context window) with MaxSim; and (v) **Hybrid RRF** fuses BM25 and Gemini dense rankings via Reciprocal Rank Fusion ($k=60$, top-100 candidates each). All architectures retrieve from the identical 2,882-node benchmark corpus introduced in §3; only the retrieval mechanism differs. Full configuration settings are in Appendix A.

4.2 Embedding Robustness

To separate architectural effects from embedding quality, we evaluate six models spanning 384–4,096 dimensions (Table 6) on the same Bengali query set under identical retrieval code, isolating embedding choice as the only varying factor.

4.3 Metrics

Retrieval (L1). We report R@1, R@5, R@10 (hereafter R@k), MRR, and nDCG@10 against gold source nodes for the 900 answerable queries. A retrieved node counts as a hit only if its exact node identifier matches the provenance-verified gold node (serving as the strict retrieval document): a deterministic criterion admitting no partial credit. For context, the random baseline R@10 for a corpus of this size is $10/2,882 \approx 0.003$.

Separation score. Beyond Recall@k, we measure how well embeddings separate the gold node from the remainder of the corpus. For query q_i with gold node g_i , let $s_{\text{gold}} = \cos(E(q_i), E(g_i))$ and s_{neg} be the mean cosine similarity between $E(q_i)$ and all 2,881 non-gold nodes. The separation score is the mean difference ($s_{\text{gold}} - s_{\text{neg}}$) across all N evaluated queries. A score of zero indicates random ranking behaviour; negative values indicate the gold node ranks below the corpus average.

4.4 Statistical Validation

We compare retrieval architectures using BCa bootstrap confidence intervals (10,000 resamples). Pairwise significance is assessed via paired Wilcoxon signed-rank tests with Holm-Bonferroni correction across 10 architecture pairs in Table 3; per-category and per-embedding tables are reported descriptively. Significance levels ($p < 0.001$ for BM25 vs. Dense; $p < 0.01$ for Hybrid RRF) and 95% CIs are reported inline in Section 5.1.

5 Results

Each subsection answers one research question using the 900 answerable Bengali queries unless otherwise noted. All reported numbers are from the post-audit verification run; Section 6 discusses mechanistic explanations.

5.1 RQ1: Benchmark Difficulty and Architecture Comparison

Table 3 presents retrieval accuracy across 900 answerable Bengali queries (random baseline R@10 $\approx$ 0.003). Hybrid RRF achieves the highest overall accuracy (R@10=0.539); lexical and dense signals provide complementary coverage. Among single-method retrievers, BM25 leads (R@10=0.506, 95% CI: [0.474, 0.538]), outperforming dense Gemini-001 (0.464, 95% CI: [0.432, 0.497]) by 9% ($p < 0.001$, Wilcoxon signed-rank test). Late-interaction ColBERT (BGE-M3, 512 tokens) reaches R@10=0.487, outperforming both single-vector dense variants (0.464, 0.408). Broken out by query category (Table 5), ColBERT is the best single architecture on KG-grounded queries (R@10=0.600, vs. Dense 0.489,

Table 3: L1 Retrieval on 900 Answerable Bengali Queries (BN $\rightarrow$ BN). Bold = best overall; underline = best single-method.

Architecture	R@1	R@5	R@10	MRR	nDCG@10
Hybrid RRF (Gemini+BM25)	.291	.466	.539	.551	.461
BM25 (Sparse)	.205	.405	<u>.506</u>	.481	.407
Dense (Gemini-001)	.320	<u>.431</u>	.464	<u>.514</u>	<u>.431</u>
Dense (BGE-M3 Native)	.281	.369	.408	.432	.370
ColBERT (BGE-M3) [‡]	.255	.414	.487	.324	.416

[‡] ColBERT evaluated at max_seq_length=512, matching the dense baselines’ context window.

Table 4: Cross-Lingual Comparison (95% BCa CIs for R@10 in brackets; 900 queries).

Setting	Dense (Gemini)	BM25	Winner
Bengali (BN $\rightarrow$ BN)	.464 [.43, .50]	.506 [.47, .54]	BM25 (+9%)
Cross-lingual (EN $\rightarrow$ BN)	.425 [.39, .46]	.004 [.00, .01]	Dense ($\approx$ 100 $\times$)
English (EN $\rightarrow$ EN)	.442 [.41, .47]	.384 [.35, .42]	Dense (+15%)

BM25 0.478) and substantially closes the gap on farmer queries relative to Dense (0.210 vs. 0.093).

5.2 RQ2: Is architectural failure determined by language mismatch?

Table 4 examines whether language boundary crossing changes retrieval behaviour independent of architecture. Within the same language, BM25 leads in Bengali (0.506 vs. 0.464) while Dense leads in English (0.442 vs. 0.384): neither architecture is universally dominant. The results diverge markedly in the cross-lingual setting (English queries $\rightarrow$ Bengali corpus), where BM25’s exact lexical matching declines sharply across script boundaries (R@10: 0.506 $\rightarrow$ 0.004, a 99% drop) and becomes effectively unusable. Multilingual dense embeddings, by contrast, largely preserve the script boundary (R@10: 0.464 $\rightarrow$ 0.425, only an 8% drop). Architecture choice cannot be separated from the language scenario it is deployed in.

5.3 RQ3: Does the query type moderate retrieval performance?

Dense retrieval’s aggregate Bengali score masks a pronounced per-category interaction (Table 5). Dense retrieval (Gemini) nearly fails on colloquial farmer queries (R@10=0.093) but reaches very high accuracy on formal safety queries (R@10=0.970), separated only by query register. Safety queries function as near-direct entity lookups, with little inferential gap to close between query and gold document (§6.1); farmer queries instead require bridging the symptom-to-entity register gap (§6.1), which explains

Table 5: R@10 by query category (95% BCa CIs in brackets; 900 verified queries; Farmer=300, Safety=200, KG-grounded=400).

Architecture	Farmer (colloquial)	Safety (formal)	KG-Ground. (formal)	Farm.+Safe. (combined)
Dense (Gemini)	.093 [.06, .13]	.970 [.94, .99]	.489 [.44, .54]	.444 [.40, .49]
BM25	.523 [.47, .58]	.539 [.47, .60]	.478 [.43, .53]	.529 [.48, .57]
ColBERT (BGE-M3) [‡]	.210 [.17, .26]	.675 [.60, .74]	.600 [.55, .65]	.396 [.35, .44]

Table 6: Multi-Embedding Analysis (95% BCa CIs for R@10 in brackets; 900 queries, BN→BN). Random baseline: separation = 0.

Model	Dim	Separation	R@10
Gemini-embedding-001	3,072	+0.127	.464 [.43, .50]
BGE-M3 (native proj.)	1,024	+0.133	.408 [.38, .44]
E5-Large (multilingual)	1,024	+0.037	.284 [.26, .32]
Qwen3 Embedding 8B	4,096	+0.108	.241 [.21, .27]
MPNet (paraphrase)	768	−0.010	.009 [.00, .02]
MiniLM (paraphrase)	384	−0.039	.005 [.00, .01]

their near-failure. BM25 is comparatively stable across categories: farmer (0.523), safety (0.539), KG (0.478). On KG-grounded queries dense leads BM25 by a narrow margin (0.489 vs. 0.478); BM25’s same-language advantage over dense comes from consistent performance across colloquial and formal query types, in contrast to dense retrieval’s bimodal split.

5.4 RQ4: How do different embedding models compare?

Evaluating six embedding models spanning 384–4,096 dimensions (Table 6) shows that retrieval-specific training dominates model scale: Gemini-001 (0.464) and BGE-M3 (0.408) outperform larger models like Qwen3-8B (0.241). Notably, BGE-M3 yields the highest mean cosine separation (+0.133 vs. Gemini’s +0.127) despite ranking second on Recall@10 (.408 vs. .464); mean separation rewards models with confident average alignment, whereas top-*k* recall penalizes a long tail of complete retrieval misses. General-purpose sentence-similarity encoders (MPNet, MiniLM) fail (R@10 < 0.01), yielding negative separation scores that indicate gold nodes are ranked below corpus average.

Configuration Audit. To verify that architecture comparisons are not implementation artifacts, we audited configuration sensitivity across embedding APIs and passage granularity. Defaulting to a sentence-similarity task type instead of the asymmetric RETRIEVAL_QUERY/RETRIEVAL_DOCUMENT encoding reduced Gemini Dense Recall@10 from 0.464 to

0.063, a 7× drop. Passage granularity is a second, independent configuration axis: at the field-standard default of 128 tokens, Bengali knowledge nodes (mean ≈1,180 characters) are truncated by 95–99%, and ColBERT underperforms even the weaker dense baseline (R@10=0.376 vs. BGE-M3’s 0.408); matching the dense baselines’ 512-token context window instead lifts ColBERT to R@10=0.487 (Table 3), reversing this ranking. Both an embedding-API task type and a passage-length truncation, two independent default settings, each distorted reported architecture rankings by a wide margin. Configuration auditing must therefore precede architecture comparison in low-resource retrieval settings. Full configuration details are in Appendix A.

6 Mechanistic Analysis

This section explains the performance patterns from Section 5 through two lenses: the query-document register gap (§6.1), and failure complementarity between BM25 and dense retrieval (§6.2).

6.1 Register Gap Analysis

Token-level Jaccard similarity across all 900 query-gold pairs reveals a near-universal lexical gap: **96.4%** of queries have Jaccard < 0.10 with their gold document (mean: 0.044; maximum: 0.172). This confirms that benchmark evaluation requires semantic matching, not direct surface overlap.

However, surface overlap varies substantially across query registers: safety queries achieve a mean Jaccard of 0.055 vs. 0.034 for farmer queries (1.6× higher alignment). Verbatim entity inspection explains dense retrieval’s collapse on colloquial farmer queries (R@10=0.093): only 3.5% of gold document entity names appear verbatim in farmer queries, while 92% are entirely absent. Farmers describe observable symptoms (*“leaves turning yellow”*), whereas authoritative documents encode formal scientific entities (*Tungro virus*) (illustrated in Table 7). Safety queries use precise chemical codes and variety names that closely match document vocabulary, so dense retrievers achieve high accuracy (0.970) on this category.

6.2 Failure Complementarity and Hybrid Attribution

Hybrid RRF’s gain over either single-method retriever is explained by complementary failure patterns between BM25 and Dense retrieval at the category level (Table 5). BM25 is comparatively stable across query registers (farmer 0.523, safety 0.539, KG-grounded 0.478), while Dense is sharply bimodal (farmer 0.093, safety 0.970). Because the two architectures reach their strongest and weakest performance on

Table 7: Illustrative register gap: a farmer-anchored query and its gold node (English glosses of Bengali text in brackets).

Farmer query (colloquial): [My chickpea plants’ leaves are turning yellow and drooping, what should I do?]
Gold node title: [Chickpea crop disease control and remedies] (B7_CH2_CHHOLA_0006_N1)
Gold node entity: <i>wilt disease</i> (formal pathological term; not present in query)
Token-level Jaccard: 0.03 (below 0.10 near-universal gap threshold, §6.1)

largely different query registers, fusing their rankings recovers queries that either method alone would miss, consistent with Hybrid RRF’s overall gain over both single methods ($R@10=0.539$ vs. 0.506 and 0.464). Persistent shared difficulty concentrates in colloquial farmer queries and multi-hop KG-grounded queries, where neither lexical nor semantic matching reliably surfaces the gold document, consistent with the symptom-to-entity register mismatch identified above as the primary barrier in low-resource retrieval.

7 Discussion

7.1 Methodological Implications

Evaluating low-resource retrieval under aggregate metrics conceals critical failure modes. Across query registers, dense retrieval collapses on colloquial farmer queries ($R@10=0.093$) while saturating on formal safety queries ($R@10=0.970$), corroborating concurrent findings on stratified retrieval evaluation [10]. Across language boundaries, BM25 wins same-script Bengali while Dense wins cross-lingually. Finally, our configuration audit shows that two independent default settings, an embedding-API task type and a passage-length truncation, each distort reported architecture rankings by a wide margin; configuration auditing must therefore precede architectural evaluation [21].

7.2 Limitations and Ethics

Limitations. Evaluation is limited to agricultural advisory and to first-stage zero-shot retrievers; learned sparse models (e.g., SPLADE) and cross-encoder rerankers are left to future work. Machine-translated English queries may carry subtle lexical shifts despite 97% back-translation equivalence on a 100-query random sample (§3.5); future work should expand translation validation to the full corpus to systematically characterize terminology preservation. However, BM25’s near-total collapse under cross-lingual querying ($R@10=0.004$) reflects an expected consequence of exact lexical matching across non-cognate scripts, since

the cross-lingual condition compares English queries directly against the untranslated Bengali corpus; this result is unlikely to be a translation artifact. Image-linked nodes (Appendix C) attach a visual reference to a subset of text nodes but are not separately embedded or evaluated as a distinct retrieval modality in this work. A human audit of 200 nodes found 12 requiring correction, so a small fraction of unresolved failure queries may reflect annotation artifacts in the benchmark itself. Source documents span 1999–2024. AgRiTRUST evaluates whether a retriever surfaces the correct authoritative source document, not whether that document’s recommendations remain current; deploying any system built on this corpus for live advisory would require a separate regulatory-currency check independent of retrieval accuracy. Finally, while reliance on proprietary APIs for initial query generation poses a reproducibility risk as models deprecate, the resulting benchmark is released as a static, version-controlled JSON corpus, supporting stable retrieval evaluations over time.

Ethics and broader impact. Annotations were completed by three compensated domain-expert annotators with informed consent. The benchmark evaluates retrieval accuracy to motivate model improvement in low-resource domain advisory and contains no personally identifiable information. Given the safety-critical query subset, retrieval accuracy does not certify advisory correctness for unsupervised deployment; systems built on AgRiTRUST should retain human oversight for high-stakes recommendations.

8 Conclusion

We introduced AgRiTRUST, a provenance-grounded benchmark for Bengali agricultural retrieval built from authoritative government publications through canonical knowledge nodes, image-linked resources, and register- and language-stratified evaluation. Beyond providing a reproducible benchmark, our experiments show that retrieval behaviour depends jointly on linguistic register, language boundary, and implementation configuration rather than retrieval architecture alone. These findings provide a stronger empirical basis for evaluating future low-resource retrieval and RAG systems.

A Reproducibility Checklist

- **Data:** 284 source PDFs, 2,882 knowledge nodes, 1,000 queries (900 with verified gold-node mappings; 100 low-agreement queries released for future work). Available at <https://huggingface.co/datasets/RaiyanKhaan/AgriTrust-RAG>.
- **BM25:** Okapi ($k_1=1.5$, $b=0.75$), char-bigram + full-word tokenization.

- **Dense (Gemini):** gemini-embedding-001, 3,072-dim, asymmetric RETRIEVAL_QUERY/RETRIEVAL_DOCUMENT task types.
- **Dense (BGE-M3):** Native projection head, 1,024-dim, L_2 -normalized.
- **ColBERT:** BGE-M3 multi-vector, 512-token sequence length, MaxSim scoring.
- **Hybrid RRF:** Reciprocal Rank Fusion ($k=60$, top-100 candidates per method).
- **Audit Variants:** sentence-similarity task type; 128-token passage truncation.
- **Stats:** BCa bootstrap (10,000 resamples), paired Wilcoxon signed-rank tests with Holm-Bonferroni correction.

B Node Representation

A representative canonical knowledge node (trimmed; full schema released with benchmark code), showing the three-layer structure (natural-language content, structured facts, deterministic provenance):

```
{ "category": "disease",
  "title_bn": "[Chickpea crop disease control and remedies]",
  "summary": "[Guidance on wilt disease control via seed treatment, variety selection, and infected plant management]",
  "key_points": [
    "[Treat seeds with Provex-200 WP at 2.5-3.0 g/kg rate]",
    "[Grow resistant varieties: BARI Chickpea-5, -9, -10]"
  ],
  "structured": { "symptoms": ["wilt disease"],
    "management": ["seed treatment (Provex-200 WP)", "fungicide spray"],
    "prevention": ["use resistant varieties"] },
  "entities": { "chemicals": ["Provex-200 WP"],
    "diseases": ["wilt disease"], "crops": ["chickpea"] },
  "generation_model": "gemini-3.1-flash-lite",
  "_provenance_layer": { "node_id": "B7_CH2_CHHOLA_0006_N1",
    "publisher": "BARI",
    "source_document": "krishiProjuktiHatboi10.pdf",
    "source_pages": [112, 112] } }
```

Field values in brackets are English translations of the stored UTF-8 Bengali text.

C Image-Linked Node Structure

Nodes with a linked visual asset extend the standard schema (Appendix B) with an `image_refs` field, without introducing a separate node type:

```
{ ...
  "image_refs": [
    { "caption_en": "[Chickpea leaf showing early wilt symptoms]",
      "image_type": "diagnostic_photo" }
  ], ... }
```

All other fields (`node_id`, `category`, `title_bn`, `_provenance_layer`) are unchanged from the node shown in Appendix B.

In total, 1,022 of 2,882 nodes (35.5%) carry one or more image references. Image references have not undergone the three-annotator human audit (§3.2) applied to node text, and

we report them as a structural resource, not an independently verified evaluation asset (§7.2).

D Node Extraction Protocol

The canonical nodes in AGRiTRUST are constructed via a two-stage generative pipeline bounded by deterministic checks.

First, an automated extraction stage identifies critical entities (crops, diseases, agricultural chemicals) from the source Markdown passage and stores them in a temporary registry. The extraction prompt enforces exact surface-form matching and forbids hallucination:

```
### Strict Bounding: Extract ONLY entities explicitly mentioned in
the text. Do not infer or add external knowledge. If a category is
missing, return an empty list.
### Surface Form: Extract the verbatim string exactly as it appears
in the Bengali text.
### Categories: "crops", "diseases_and_pests", "chemicals".
### Output: Emit a valid JSON object ONLY.
```

Second, the Markdown passage is converted into structured JSON node(s) by Gemini-3.1-Flash-Lite. Following generation, a deterministic script verifies that the entities extracted in Stage 1 exactly match those present in the generated node. Provenance metadata is then injected purely deterministically. The abridged generation instruction enforces 1-to-N dynamic chunking and semantic bounding:

```
### Strict Domain Bounding: The generated node(s) must only contain
information from the specific .md document provided... Never claim
a crop or topic that the trace and body do not support.
### Faithfulness: Every factual claim must come from the section
you were given. You do not add facts, doses, chemical names, or
yields from memory. If you cannot ground a field in the text, leave
it blank.
### How many nodes: One coherent topic → one node. Several distinct
sub-topics (separate varieties, a disease and its treatment) → one
node each. Each node must be a distinct, non-redundant unit.
### Category: Pick the single best from: variety, disease,
pest, fertilizer, cultivation_practice, post_harvest, seed_tech,
machinery, irrigation, ipm, cropping_system, food_safety, general.
### YOUR OUTPUT: Emit semantic JSON payload ONLY. [ { "category":
"...", "content_bn": "...", "structured": { ... } } ]
```

References

- [1] Mohd Ruhul Ameen, Akif Islam, Farjana Aktar, and M Saifuzzaman Rafat. 2026. KrishokBondhu: A Retrieval-Augmented Voice-Based Agricultural Advisory Call Center for Bengali Farmers. In *2026 IEEE 2nd International Conference on Quantum Photonics, Artificial Intelligence & Networking (QPAIN)*. IEEE, 1–6.
- [2] David M. Eberhard, Gary F. Simons, and Charles D. Fennig. 2024. Ethnologue: Languages of the World. <https://www.ethnologue.com/language/ben/>. Online version.
- [3] Robert Friel, Masha Belyi, and Atindriyo Sanyal. 2024. Ragbench: Explainable benchmark for retrieval-augmented generation systems. *arXiv preprint arXiv:2407.11005* (2024).
- [4] George W. Furnas, Thomas K. Landauer, Louis M. Gomez, and Susan T. Dumais. 1987. The vocabulary problem in human-system communication. *Commun. ACM* 30, 11 (1987), 964–971.

- [5] Seongtae Hong, Youngjoon Jang, Jungseob Lee, Hyeonseok Moon, and Heuiseok Lim. 2026. Improving semantic proximity in information retrieval through cross-lingual alignment. In *The Fourteenth International Conference on Learning Representations*.
- [6] Md Asif Hossain, Nabil Subhan, Mantasha Rahman Mahi, and Janatul Ferdous Nabila. 2026. Cost-Efficient Cross-Lingual Retrieval-Augmented Generation for Low-Resource Languages: A Case Study in Bengali Agricultural Advisory. *arXiv preprint arXiv:2601.02065* (2026).
- [7] Tomasz Jurczyk and Jinho D. Choi. 2017. Cross-genre Document Retrieval: Matching between Conversational and Formal Writings. In *Proceedings of the First Workshop on Building Linguistically Generalizable NLP Systems*. Association for Computational Linguistics.
- [8] Mohsinul Kabir, Mohammed Saidul Islam, Md Tahmid Rahman Laskar, Mir Tafseer Nayeem, M Saiful Bari, and Enamul Hoque. 2024. BenLLM-eval: A comprehensive evaluation into the potentials and pitfalls of large language models on Bengali NLP. In *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)*. 2238–2252.
- [9] Omar Khattab and Matei Zaharia. 2020. Colbert: Efficient and effective passage search via contextualized late interaction over bert. In *Proceedings of the 43rd International ACM SIGIR conference on research and development in Information Retrieval*. 39–48.
- [10] Andrew Klearman, Radu Revutchi, Rohin Garg, Rishav Chakravarti, Samuel Marc Denton, and Yuan Xue. 2026. Coverage, Not Averages: Semantic Stratification for Trustworthy Retrieval Evaluation. *arXiv preprint arXiv:2604.20763* (2026).
- [11] Seungyoon Lee, Minhyuk Kim, Seongtae Hong, Youngjoon Jang, Dongsuk Oh, and Heui-Seok Lim. 2026. CLEAR: Cross-Lingual Enhancement in Retrieval via Reverse-training. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. 347–362.
- [12] Noshin Nawal, Sanju Basak, and Rifat Shahriyar. 2024. *Effective retrieval-augmented generation for open domain question answering in bengali*. Ph.D. Dissertation. Ph. D. dissertation.
- [13] Antoine Nzeyimana and Andre Niyongabo Rubungo. 2025. Kinya-ColBERT: A Lexically Grounded Retrieval Model for Low-Resource Retrieval-Augmented Generation. *arXiv preprint arXiv:2507.03241* (2025).
- [14] Nishat Raihan and Marcos Zampieri. 2025. TigerLLM-a family of Bangla large language models. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*. 887–896.
- [15] Khan Raiyan Ibne Reza, Sumaiya Tabassum Nimi, and Omar-Ibne Shahid. 2026. KrishokChat: A Provenance-Traceable Multi-Task Bengali Agricultural Benchmark with Safety-Critical Chemical Advisory. In *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics*.
- [16] Stephen Robertson and Hugo Zaragoza. 2009. *The probabilistic relevance framework: BM25 and beyond*. Vol. 4. Now Publishers Inc.
- [17] Namita Singh, Jacqueline Wang’ombe, Nereah Okanga, Tetyana Zelenska, Jona Repishti, Jayasankar G K, Sanjeev Mishra, Rajsekar Manokaran, Vineet Singh, Mohammed Irfan Rafiq, Rikin Gandhi, and Akshay Nambi. 2024. Farmer.Chat: Scaling AI-Powered Agricultural Services for Smallholder Farmers. *arXiv preprint arXiv:2409.08916* (2024).
- [18] Ionut Teodor Sorodoc, Leonardo FR Ribeiro, Rexhina Blloshmi, Christopher Davis, and Adrià de Gispert. 2025. Garage: A benchmark with grounding annotations for rag evaluation. In *Findings of the Association for Computational Linguistics: ACL 2025*. 17030–17049.
- [19] Jan Strich, Enes Kutay Isgorur, Maximilian Trescher, Chris Biemann, and Martin Semmann. 2026. T2-RAGBench: Text-and-Table Aware Retrieval-Augmented Generation. In *Proceedings of the 19th Conference of the European Chapter of the Association for Computational Linguistics (Volume 1: Long Papers)*. 165–191.
- [20] Nandan Thakur, Nils Reimers, Andreas Rücklé, Abhishek Srivastava, and Iryna Gurevych. 2021. Beir: A heterogenous benchmark for zero-shot evaluation of information retrieval models. *arXiv preprint arXiv:2104.08663* (2021).
- [21] Liang Wang, Nan Yang, Xiaolong Huang, Linjun Yang, Rangan Majumder, and Furu Wei. 2024. Multilingual E5 Text Embeddings: A Technical Report. *arXiv preprint arXiv:2402.05672* (2024).
- [22] Jie Wu, Zhaochun Ren, and Suzan Verberne. 2024. What are the limits of cross-lingual dense passage retrieval for low-resource languages? *arXiv preprint arXiv:2408.11942* (2024).
- [23] Xiao Yang, Kai Sun, Hao Xin, Yushi Sun, Nikita Bhalla, Xiangsen Chen, Sajal Choudhary, Rongze D Gui, Ziran W Jiang, Ziyu Jiang, et al. 2024. Crag-comprehensive rag benchmark. *Advances in Neural Information Processing Systems* 37 (2024), 10470–10490.
- [24] Tilahun Yeshambel, Moncef Garouani, Serge Molina, and Josiane Mothe. 2025. Dense Retrieval for Low Resource languages-the Case of Amharic Language. In *Proceedings of the 48th International ACM SIGIR Conference on Research and Development in Information Retrieval*. 3098–3100.
- [25] Qing T. Zeng and Tony Tse. 2006. Exploring and developing consumer health vocabularies. *Journal of the American Medical Informatics Association* 13, 1 (2006), 24–29.
- [26] Xinyu Zhang, Xueguang Ma, Peng Shi, and Jimmy Lin. 2021. Mr. TyDi: A multi-lingual benchmark for dense retrieval. In *Proceedings of the 1st workshop on multilingual representation learning*. 127–137.
- [27] Xinyu Zhang, Nandan Thakur, Odunayo Ogundepo, Ehsan Kamalloo, David Alfonso-Hermelo, Xiaoguang Li, Qun Liu, Mehdi Rezagholizadeh, and Jimmy Lin. 2023. Miracl: A multilingual retrieval dataset covering 18 diverse languages. *Transactions of the Association for Computational Linguistics* 11 (2023), 1114–1131.